Yunfei **Luo**

Ph.D. Student at Halıcıoğlu School of Data Science and Computing, University of California San Diego

(413) 404 2124 @ yul268@ucsd.edu San Diego, CA
yunfeiluo.github.io linkedin.com/in/yunfei-luo github.com/yunfeiluo

Reseach Interests

General AI/ML, Digital Health, Signal Processing, Multimodal Learning, Multitask Learning, Foundation Model, Reinforcement Learning

Specific Yunfei's research focuses on machine learning in the field of digital health, signal processing, multimodal modeling, and multitask learning. He aims to develop a robust and well-generalized methodology that enhances intelligent monitoring, supports timely interventions, and is deployable on edge devices.

Applications Mental Health, Substance Usage, Kidney Dialysis, Digital Health Monitoring

Education

09.2023 - present
GPA : $3.60/4.0$

Ph.D. Student in Data Science, AI/ML/Digital-Health
Halıcıoğlu School of Data Science and Computing (**HSDSC**), University of California (**UC San Diego**)
- Advisor : Tauhidur Rahman
- Research : Machine Learning, Multimodal Learning, Digital Health, Signal Processing

09.2021 - 05.2023
GPA : $3.92/4.0$

Master of Science in Computer Science
Manning College of Information and Computer Science (**CICS**), University of Massachusetts Amherst (**UMass Amherst**)
- Advisor : Madalina Fiterau Brostean | Information Fusion Lab
- Courses Taken : Advanced Machine Learning, Neural Networks, Probabilistic Graphical Models, Reinforcement Learning, Advanced Natural Language Processing, Advanced IoT

09.2017 - 05.2021
GPA : $3.84/4.0$

Bachelor of Science in Computer Science | **Bachelor of Science in Mathematics** (Dual Degrees)
CICS | College of Natural Sciences (CNS), UMass Amherst
- Machine Learning and Computing Concentration

Fellowships, Scholarships, Awards, and Honors

2023 - 2024 ● Institute Fellowship | UC San Diego, HSDSC

2021 - 2023 ● Bay State Fellowship | Full Tuition Scholarship, UMass Amherst, CICS

2017 - 2021 ● Outstanding Undergrad Course Assistant Award, UMass Amherst, CICS
● Chancellor's Award, UMass Amherst

Teaching Experience

09.2021 to 06.2023

Graduate Teaching Assistant, CICS, UMass Amherst
- CS 311 Introduction to Algorithm in Fall 2021 and Spring 2023
- CS 590OP Applied Numerical Optimization, and CS 108 Foundation of Data Science in Fall 2022
- CS 220 Programming Methodology in Spring 2022
- Held office hours, led weekly discussion sessions, graded course works, answered questions on course forum, attended weekly TA meeting for discussing the progress/improvements of the course

Javascript | Algorithms | Numerical Optimization | Data Science | Teaching

01.2020 to 06.2021

Undergraduate Course Assistant, CICS, UMass Amherst

- CS 240 Reasoning under Uncertainty in Spring 2021, CS 311 Introduction to Algorithms in Fall 2020, CS 220 Programming Methodology in Spring 2020
- Helped with weekly discussion session, graded course works, and answered questions on Piazza

Javascript | Algorithms | Probability | Teaching

Work Experience

07.2023 to present

Graduate Student Researcher, AI/ML/Digital-Health, HSDSC, UC San Diego

- Research focus on machine learning in signal processing and digital health. Detailed works are presented in the *Publications* section
- Teaching Assistant : DSC-80 The Practice and Application of Data Science in Fall 2026

Research and Development | Machine Learning | Digital Health | Signal Processing | Natural Language Processing

06.2026 to 09.2026

Data Science Research Intern, R&D, Verily Health Inc.

- Research and Development in Machine Learning and Digital Health

Research and Development | Machine Learning | Digital Health | Signal Processing | Foundation Model

06.2022 to 08.2022

Machine Learning Engineering Intern, Center for Data Science, UMass Amherst

- Developed platform for Building Segmentation and Damage Assessment based on Satellite Imagery
- Integrated the Machine Learning pipeline along with User Interface for Imagery Fetching and Labeling
- Conducted Research and Experiments with various Finetuning setting and Machine Learning techniques to support selections of final-deployed hyper-parameters and models
- Delivered a presentation of the project to the Red Cross (funder and client), resulting in exceptional satisfaction and securing an additional year of funding for the initiative
- Worked as Internship in Data Science for Common Good program during the summer

Computer Vision | Machine Learning | Disaster Response | Software Development | Full-Stack Development

06.2020 to 05.2021

Undergraduate Reseach Assistant, Institute for Applied Life Sciences, UMass Amherst

- Built pipelines for fMRI data preprocessing and statistical analysis
- Conducted Research and Experiments with Machine Learning models for Stroke Detection : Autoencoders, CNNs, and GANs
- Worked as Internship in Core Summer Intern program, and as Undergraduate Research Assistant after summer internship

Machine Learning | fMRI | Data Preparation | Data Engineering

Publications

2026 Published

Toward Foundation Model for Multivariate Wearable Sensing of Physiological Signals, ACM Computing for Healthcare (ACM HEALTH), Journal

Yunfei Luo, Yuliang Chen, Asif Salekin, Tauhidur Rahman

Signal Processing | Foundation Model | Natural Language Processing | Machine Learning | Neural Networks | Digital Health

2026 Published

Personalized entropy-informed deep learning for identifying opioid misuse, Nature Mental Health, Journal

Yunfei Luo, Iman Deznabi, Bhanu Gullapalli, Mark Tuomenoksa, Eric Garland, Madalina Fiterau, Tauhidur Rahman

Signal Processing | Natural Language Processing | Heart Rate Variability | Affective States | Opioid | Clinical Treatment | Machine Learning | Multimodal Fusion | Digital Health

2026 Published

Transformer-Based Deep Learning Model for Predicting Hemoglobin Response to Mircera® in Hemodialysis Patients, Blood Purification, Journal

Siwei Zhao, **Yunfei Luo**, Joseph Mahaffy, Subhasis Dasgupta, Armin Ahmadi, Jerome Tannenbaum, Tauhidur Rahman, Rakesh Malhotra

Clinical Treatment | Machine Learning | Digital Health

2026 | Automated Dysphagia Screening Using Noninvasive Neck Acoustic Sensing, IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), Conference
Published | Jade Chng, Rong Xing (Co-1st Author), **Yunfei Luo (Co-1st Author)**, Kristen Linnemeyer-Risser, Tauhidur Rahman, Andrew Yousef, Philip A Weissbrod
Signal Processing | Acoustic Signal | Data Collection | Clinical Treatment | Applied Machine Learning | Digital Health

2025 | Sounding out Dysphagia : Using a Digital Stethoscope to Assess Audiometric Differences in Swallowing, The Laryngoscope, Journal
Published | Andrew Yousef, Sacha Moufarrej, David Zeng, Bao Luu, Rong Xing, James S. Crook, **Yunfei Luo**, Kristen Linnemeyer-Risser, Tauhidur Rahman, Philip A Weissbrod
Signal Processing | Digital Health | Dysphagia | Acoustics | Non-invasive

2025 | Opioid Misuse Detection from Cognitive and Physiological Data with Temporal Fusion Deep Learning, Drug and Alcohol Dependence, Journal
Published | Bhanu Gullapalli, **Yunfei Luo**, Tauhidur Rahman, Eric L. Garland
Signal Processing | Machine Learning | Digital Health | Addiction | Psychophysiology | Chronic Pain

2024 | Dynamic Clustering via Branched Deep Learning Enhances Personalization of Stress Prediction from Mobile Sensor Data, Nature Scientific Report, Journal
Published | **Yunfei Luo**, Iman Deznabi, Abhinav Shaw, Tauhidur Rahman, Madalina Fiterau Brostean
Multi-modal | Fusion Methods | Mental Health | Machine Learning | Neural Networks | Time Series

2024 | Start Simple : Progressive Difficulty Multitask Learning, The Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL) 2024 SRW Proceedings, Conference
Published | **Yunfei Luo**, Yuyang Liu, Rukai Cai, Tauhidur Rahman
Natural Language Processing | Multi-task Learning | Machine Learning | Neural Networks

2023 | Agent Performing Autonomous Stock Trading under Good and Bad Situations, AI for Agent-Based Modelling (AI4ABM) in International Conference on Learning Representations (ICLR) 2023, Workshop
Published | **Yunfei Luo**, Zhangqi Duan
Reinforcement Learning | Machine Learning | Neural Networks

Papers Under Preparation

2026 | Toward World Modeling of Physiological Signals with Chaos-Theoretic Balancing and Latent Dynamics, -, Conference
Under Revision | **Yunfei Luo**, Xi Chen, Yuliang Chen, Lanshuang Zhang, Md Mofijul Islam, Siwei Zhao, Peter Kotanko, Subhasis Dasgupta, Andrew Campbell, Rakesh Malhotra, Tauhidur Rahman
Signal Processing | Foundation Model | World Model | Machine Learning | Neural Networks | Digital Health

2026 | Simulate before Acting : Counterfactual Exploration-Driven Reinforcement Learning for Acute Intervention in Intradialytic Hypotension, -, Journal
Under Review | **Yunfei Luo**, Siwei Zhao, Subhasis Dasgupta, Joseph M Mahaffy, Peter Kotanko, Rakesh Malhotra, Tauhidur Rahman
Signal Processing | Reinforcement Learning | Nephrology | Clinical Treatment | Machine Learning | Digital Health

2026 | Real Time Intradialytic Hypotension Forecasting Leveraging Modern Deep Learning Methods with Multimodal Fusion of Physiological Data and Electronic Health Records, npj Digital Medicine, Journal
Await Decision | **Yunfei Luo**, Siwei Zhao, Subhasis Dasgupta, Joseph M Mahaffy, Peter Kotanko, Tauhidur Rahman (Equal Senior Author), Rakesh Malhotra (Equal Senior Author)
Signal Processing | Nephrology | Clinical Treatment | Machine Learning | Multimodal Fusion | Digital Health

Patents

2025 | US2025/016369 - System and Method for Personalized Closed-Loop Opioid Addiction Management with Mobile and Wearable Sensing of Administrations, Affective States, and Misuse Risk Scores (**Full US Utility patent application filed on Feb 18, 2025**)

Voluntary Professional Activities

Talks

- "Toward Foundation Model for Multivariate Wearable Sensing of Physiological Signals" ACM HEALTH panel at **IEEE/ACM conference on Connected Health : Applications, Systems and Engineering Technologies (CHASE)**, August 4th 2026
- "Toward Foundation Model for Multivariate Wearable Sensing of Physiological Signals", Lightning Talk at **Conference of Naval Applications of Machine Learning (NAML)**, March 3rd 2026
- "Foundation Model for Multivariate Wearable Sensing of Physiological Signals", **Google Research Journal Club**, May 27th 2025
- "Health AI, foundation models, agents, and related research", **Dexcom Journal Club**, April 18th 2023
- "Integrating Machine Learning Models to Your Applications" , **Workshop in SackHacks V**, November 4th 2023

Reviewer

- Association for the Advancement of Artificial Intelligence (**AAAI**)
  - Main Track 2027 (Program Committee (PC) Member)
- International Conference on Learning Representations (**ICLR**)
  - Main Track 2026
  - Workshop on Time Series in the Age of Large Models, International Conference on Learning Representations (TSALM), 2026
  - Workshop on AI with Recursive Self-Improvement (RSI), 2026
- Neural Information Processing Systems (**NeurIPS**)
  - Main Track 2026
- International Conference on Machine Learning (**ICML**)
  - Main Track 2026
  - Workshop on Foundation Models for Structured Data, International Conference on Machine Learning (FMSD), 2025, 2026
- Conference on Computational Natural Language Learning (**CoNLL**)
  - Main Track 2024 to 2026
- **npj Digital Medicine**
  - Journal Paper in 2026
- Proceedings of ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (**IMWUT**)
  - Journal Papers in 2024 to 2026

Professional Certifications

2022	IBM Artificial Intelligence Engineering Professional Certificate
2022	Neuroscience and Neuroimaging Specialization Certificate
2020-2023	TensorFlow Developer Professional Certificate

References

Tauhidur Rahman
Assistant Professor of Data Science
University of California San Diego
Director of MOSAIC Lab
@ trahman@ucsd.edu

Benjamin Smarr
Associate Professor of Data Science
University of California San Diego
Department of Bioengineering
@ bsmarr@ucsd.edu

Tom Bernardin
Data Scientist
University of Massachusetts Amherst
Director of the Center for Data Science
@ tbernard@cs.umass.edu

Rakesh Malhotra
Assistant Professor of Medicine
University of California San Diego
Division of Nephrology-Hypertension
@ r3malhotra@health.ucsd.edu

Madalina Fiterau Brostean
Assistant Professor of Computer Science
University of Massachusetts Amherst
Lead of the Information Fusion Lab
@ mfiterau@cs.umass.edu

Daniel Sheldon
Associate Professor of Computer Science
University of Massachusetts Amherst
College of Information and Computer Science
@ sheldon@cs.umass.edu

Eric L. Garland
Professor in the Department of Psychiatry
University of California San Diego
Director of UCSD ONEMIND
@ egarland@health.ucsd.edu

Subhransu Maji
Associate Professor of Computer Science
University of Massachusetts Amherst
Co-Director of Computer Vision Lab
@ smaji@cs.umass.edu